Shiok POS

Software Architecture & UML Documentation • v1.3.0

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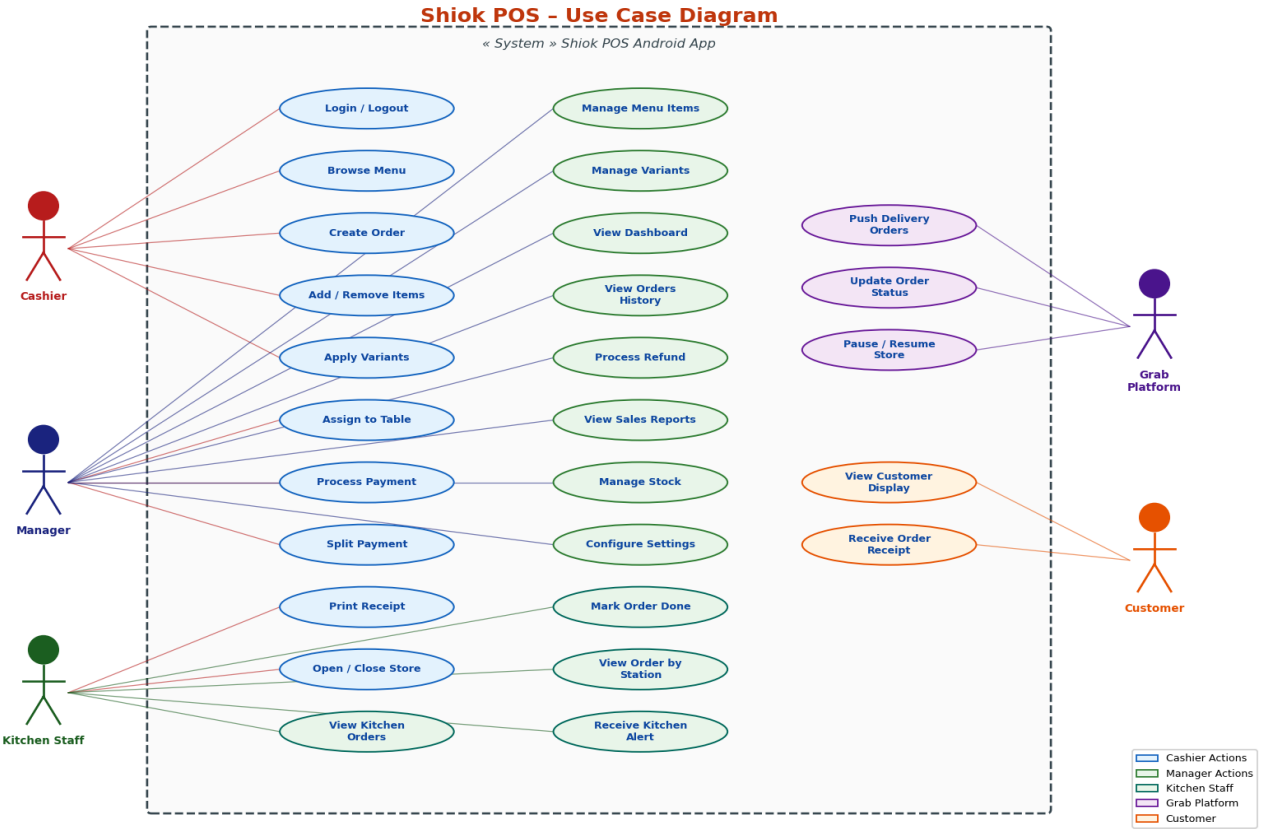
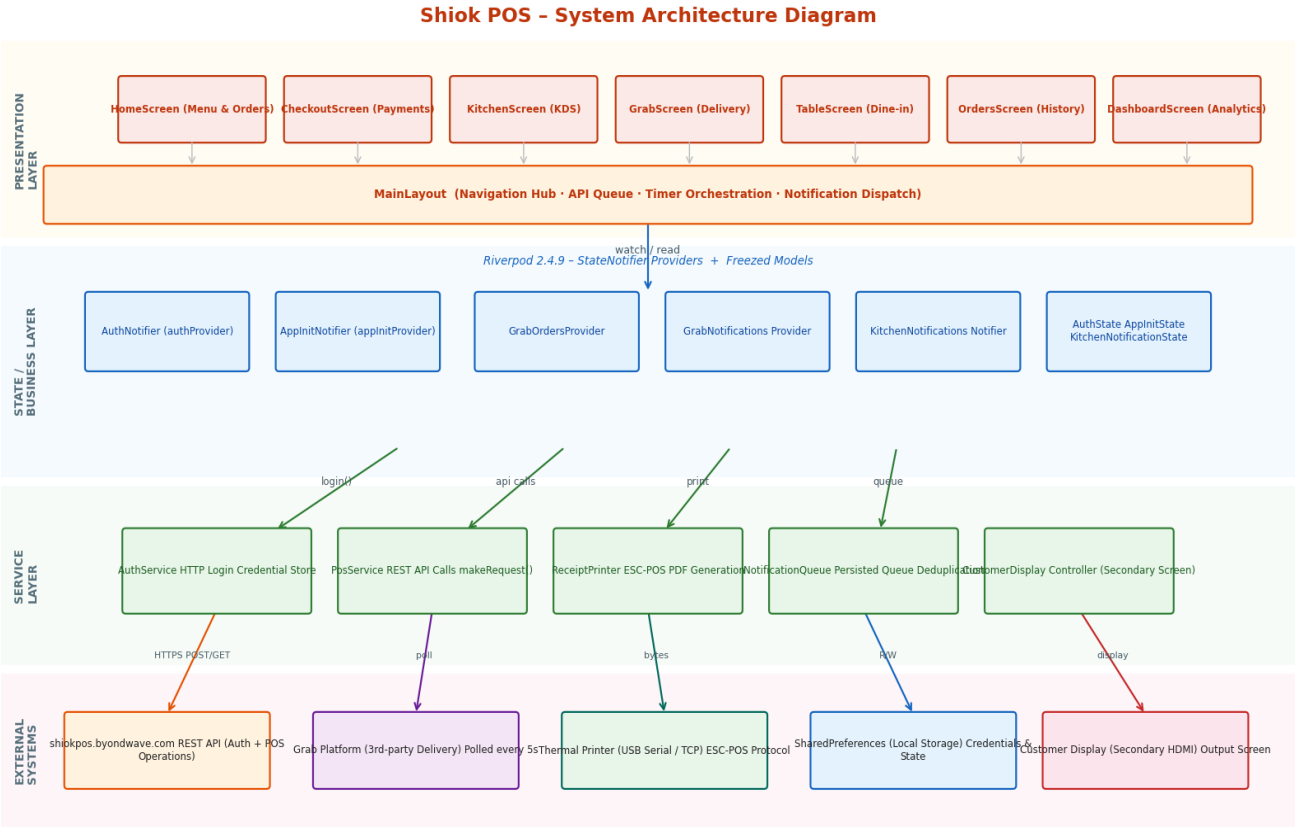
1. Use Case Diagram

Shows all system actors and the actions (use cases) they can perform within the Shiok POS boundary.



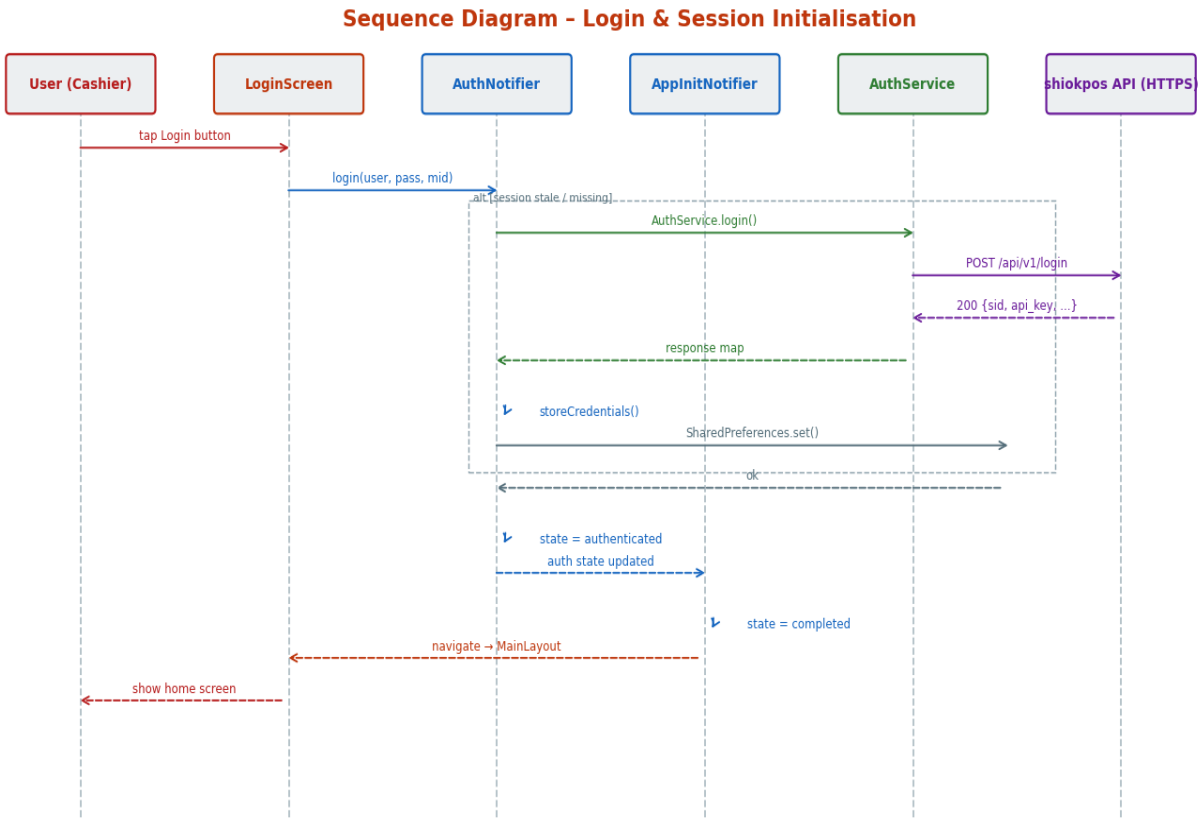
2. System Architecture Diagram

Layered view of the application from Presentation through State/Business, Service, and External Systems.



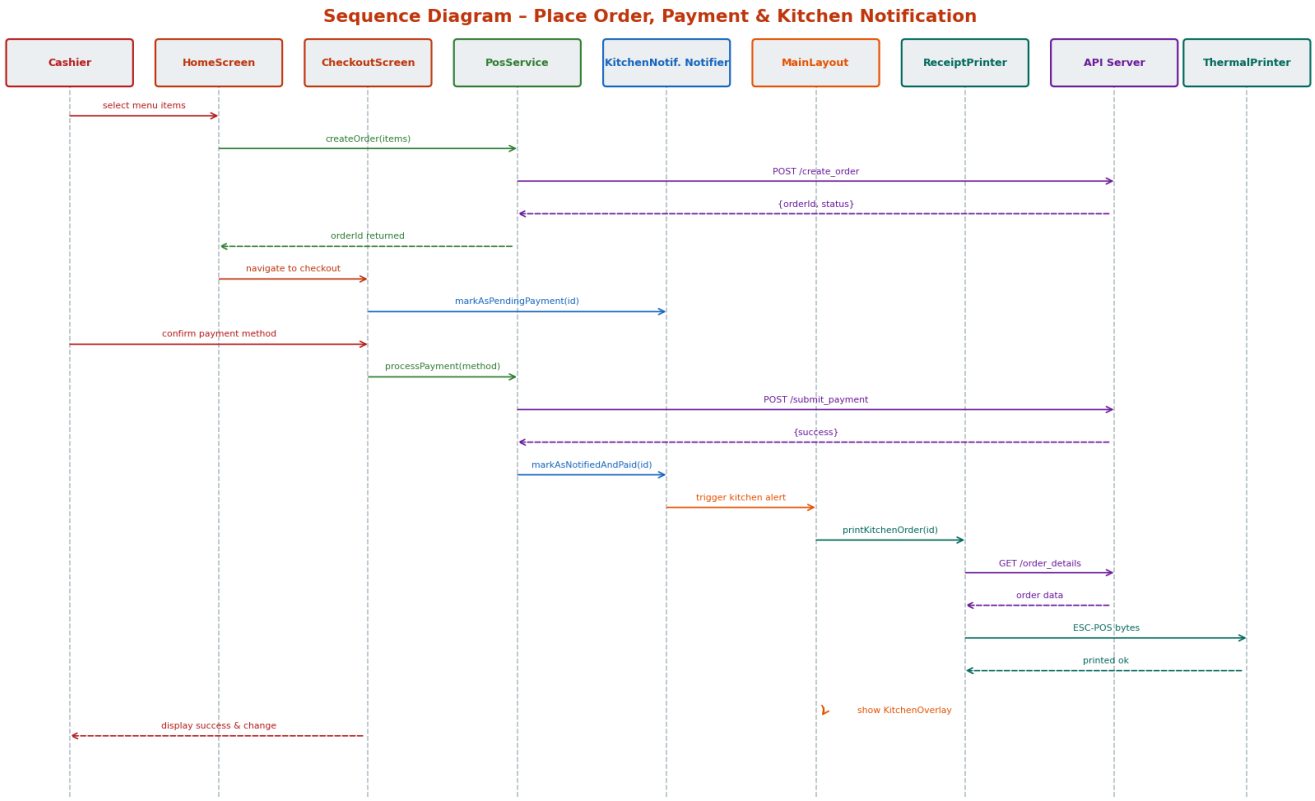
3a. Sequence Diagram – Login & Session Initialisation

Traces the full flow from the user tapping Login through AuthService, the remote API, credential persistence, and final navigation to MainLayout.



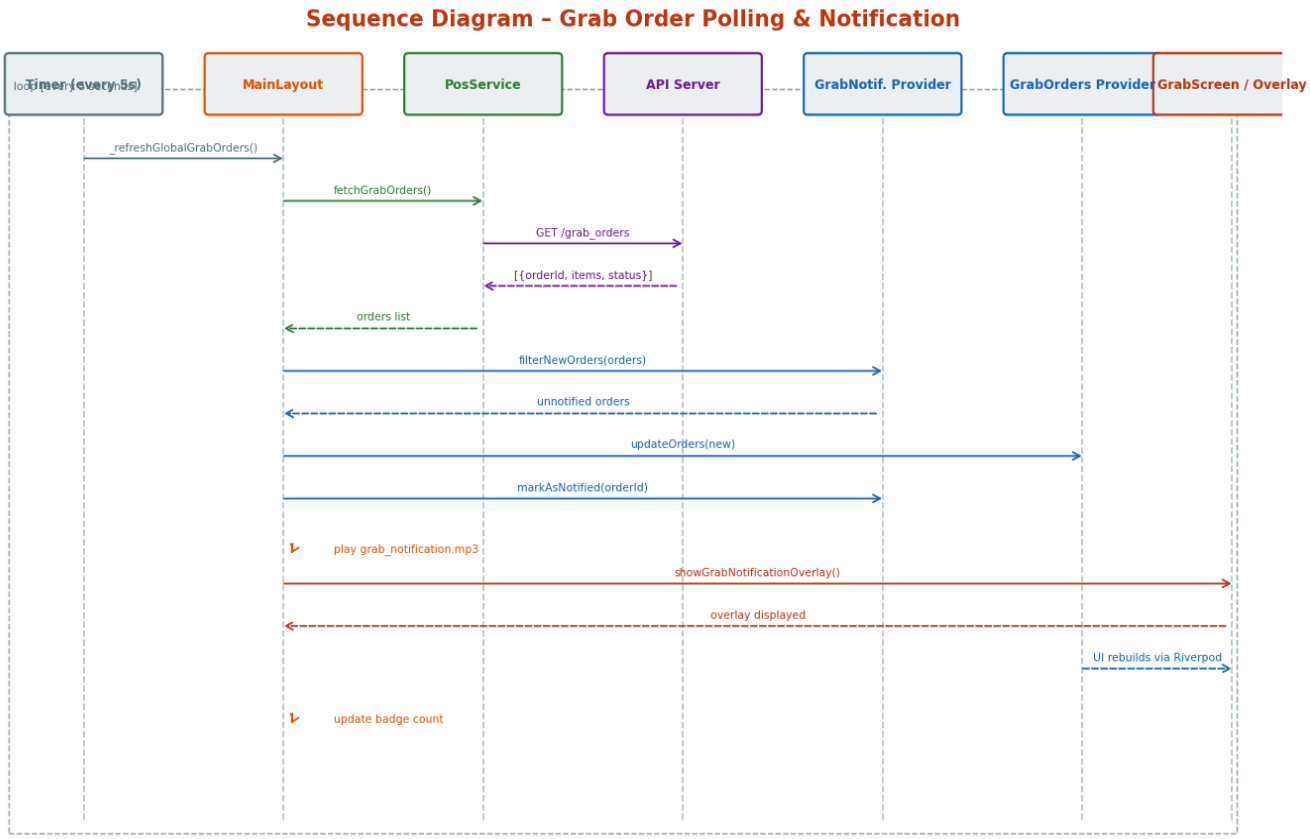
3b. Sequence Diagram – Place Order, Payment & Kitchen Notification

End-to-end flow from item selection → order creation → checkout → payment → kitchen ESC-POS print.



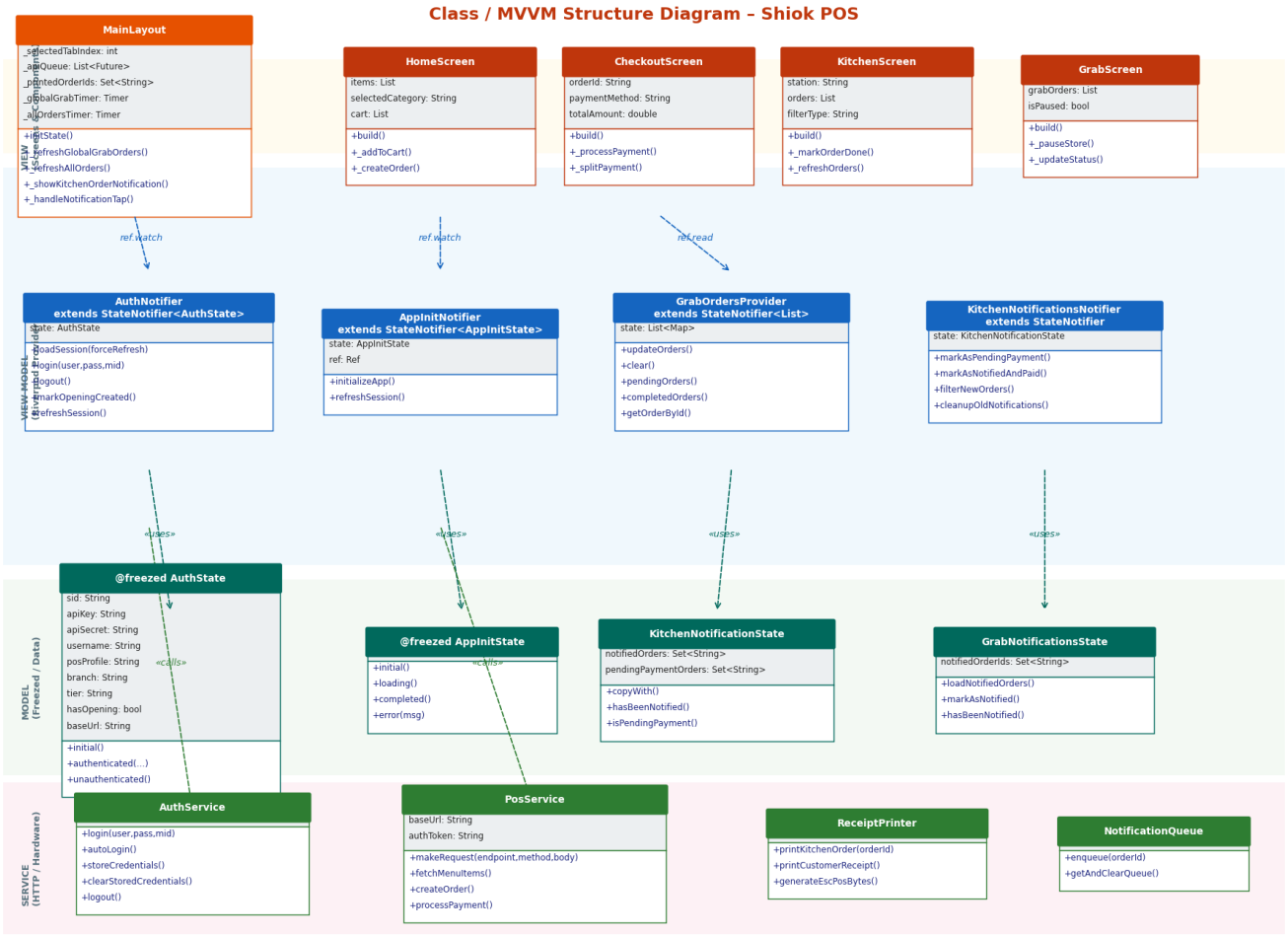
3c. Sequence Diagram – Grab Order Polling

Shows the 5-second polling loop that fetches Grab delivery orders, deduplicates them, and triggers overlay notifications with audio.



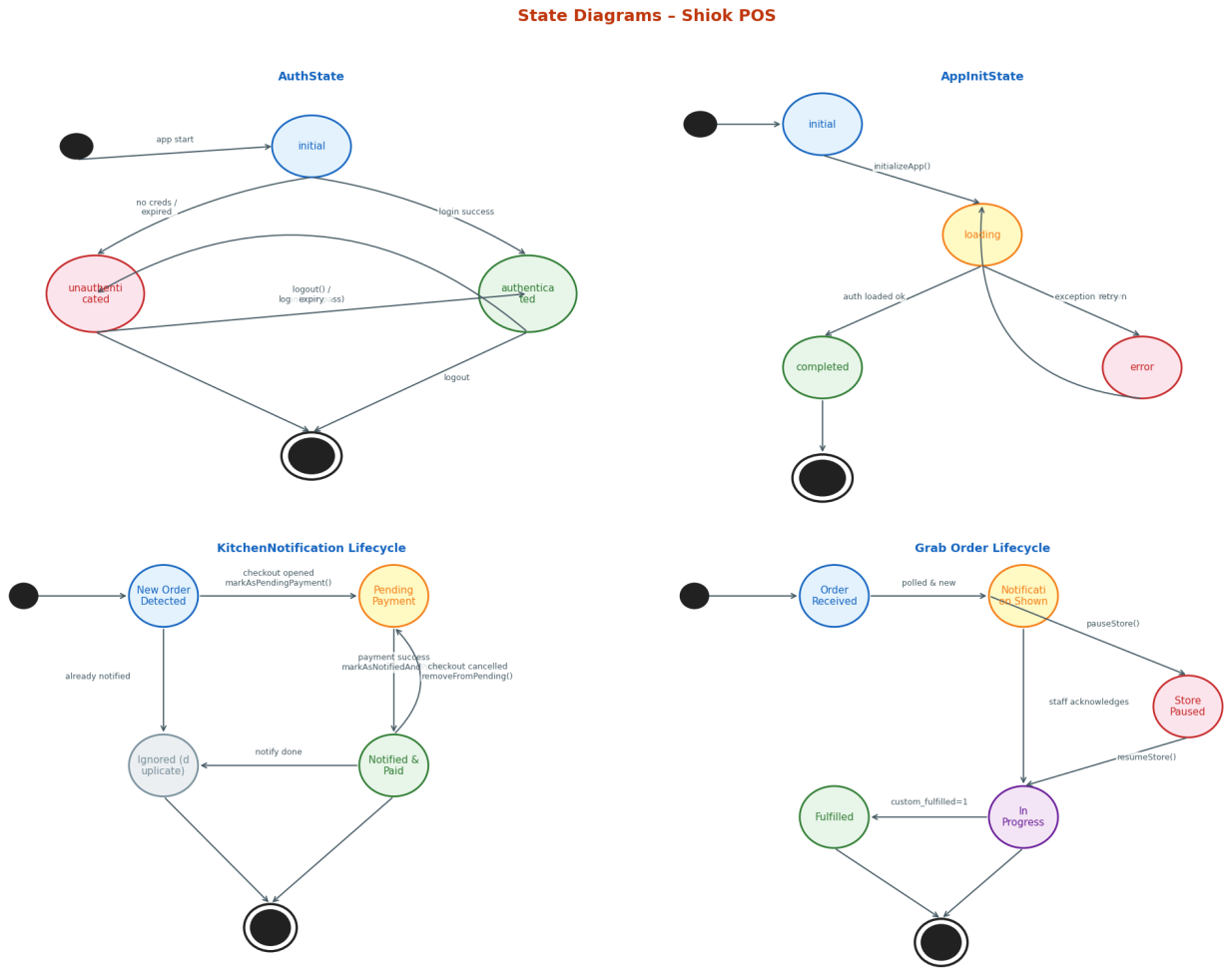
4. Class / MVVM Structure Diagram

All key classes grouped into MVVM layers: View (screens), ViewModel (Riverpod notifiers), Model (Freezed), and Service (HTTP/hardware).



5. State Diagrams

Four independent state machines governing the core runtime behaviour of the application.



6. Enhancement Suggestions

SECURITY

Enhancement	Priority
Replace SharedPreferences with flutter_secure_storage — Passwords are currently stored in plain SharedPreferences (noted in auth_service.dart line 31). Migrate to Keystore/Keychain-backed secure storage to prevent extraction on rooted devices.	Critical
Remove plaintext password storage — The password key in SharedPreferences is unnecessary for session renewal if a proper token-refresh endpoint exists. Store only the refresh token, not the raw password.	Critical
Certificate pinning — API calls go to shiokpos.byondwave.com over HTTPS. Adding cert pinning via http_certificate_pinning or the dio package prevents MitM attacks on the merchant's local network.	High
Obfuscate default API base URL — The fallback URL "https://asdf.byondwave.com" is a development artefact that leaks server topology; it should be replaced with a production default or loaded remotely.	Medium

ARCHITECTURE & CODE QUALITY

Enhancement	Priority
Extract API queue to a dedicated service — The _apiQueue logic lives inside MainLayout (a UI widget). Moving it to a PosQueueService Riverpod provider will improve testability and separate concerns.	High
Standardise error handling — PosService.makeRequest() returns raw maps; a Result sealed class (achievable with freezed) would eliminate scattered null-checks and make error paths explicit.	High
Repository pattern for API calls — PosService currently mixes HTTP transport with business rules. Splitting into a PosRepository (data) and PosUseCase (logic) layer will make unit testing straightforward.	Medium
Remove legacy markAsNotified() — KitchenNotificationsNotifier already marks it as a legacy method. Removing it and enforcing markAsNotifiedAndPaid() will eliminate the dual-path risk.	Low

PERFORMANCE & RELIABILITY

Enhancement	Priority
Adaptive polling intervals — Both grab and kitchen timers fire every 5 s unconditionally. Implementing exponential back-off when the app is backgrounded or the network is unavailable will reduce battery drain on idle devices.	High
Offline-first order queue — If the POS loses connectivity mid-order, there is no local queue. Storing pending orders in SQLite (via drift or sqflite) and syncing when reconnected would prevent lost sales.	High
Clean up KitchenNotification storage — The current cleanup shuffles and truncates randomly when > 1 000 IDs accumulate. A time-based LRU eviction (store creation timestamp alongside ID) is more predictable.	Medium
Paginate menu items on HomeScreen — Large menus load all items at once. Lazy loading or virtual scrolling will keep the UI responsive as the item catalogue grows.	Medium

TESTING

Enhancement	Priority
Unit tests for all Notifiers — AuthNotifier, AppInitNotifier, and KitchenNotificationsNotifier have no test coverage. Riverpod's ProviderContainer makes isolated tests trivial.	High

Enhancement	Priority
Integration test for payment flow — The checkout → payment → kitchen notification path is business-critical. A flutter_test golden test or an integration_test suite covering this end-to-end would catch regressions immediately.	High
Mock PosService in widget tests — Screens directly call PosService. Injecting it via a Riverpod override in tests would allow fast, hermetic widget tests without hitting the real API.	Medium

UX / FEATURES

Enhancement	Priority
Biometric authentication — Adding local_auth for fingerprint/face unlock would speed up re-login after the 4-hour session stale timeout.	Medium
Offline receipt preview — Cashiers should be able to reprint or preview a receipt without network connectivity by caching the last N receipts locally.	Medium
Push notifications via FCM — Replacing the 5-second Grab polling loop with Firebase Cloud Messaging push notifications would reduce latency and battery usage significantly.	High
Dashboard date-range picker for analytics — DashboardScreen currently uses a fixed date selector. A proper date-range calendar picker would make the reporting screen more useful for managers.	Low
Role-based access control — Tier flags gate some features, but there is no per-user permission model. Adding roles (Cashier / Manager / Admin) with visible tab restrictions would improve security and UX.	Medium